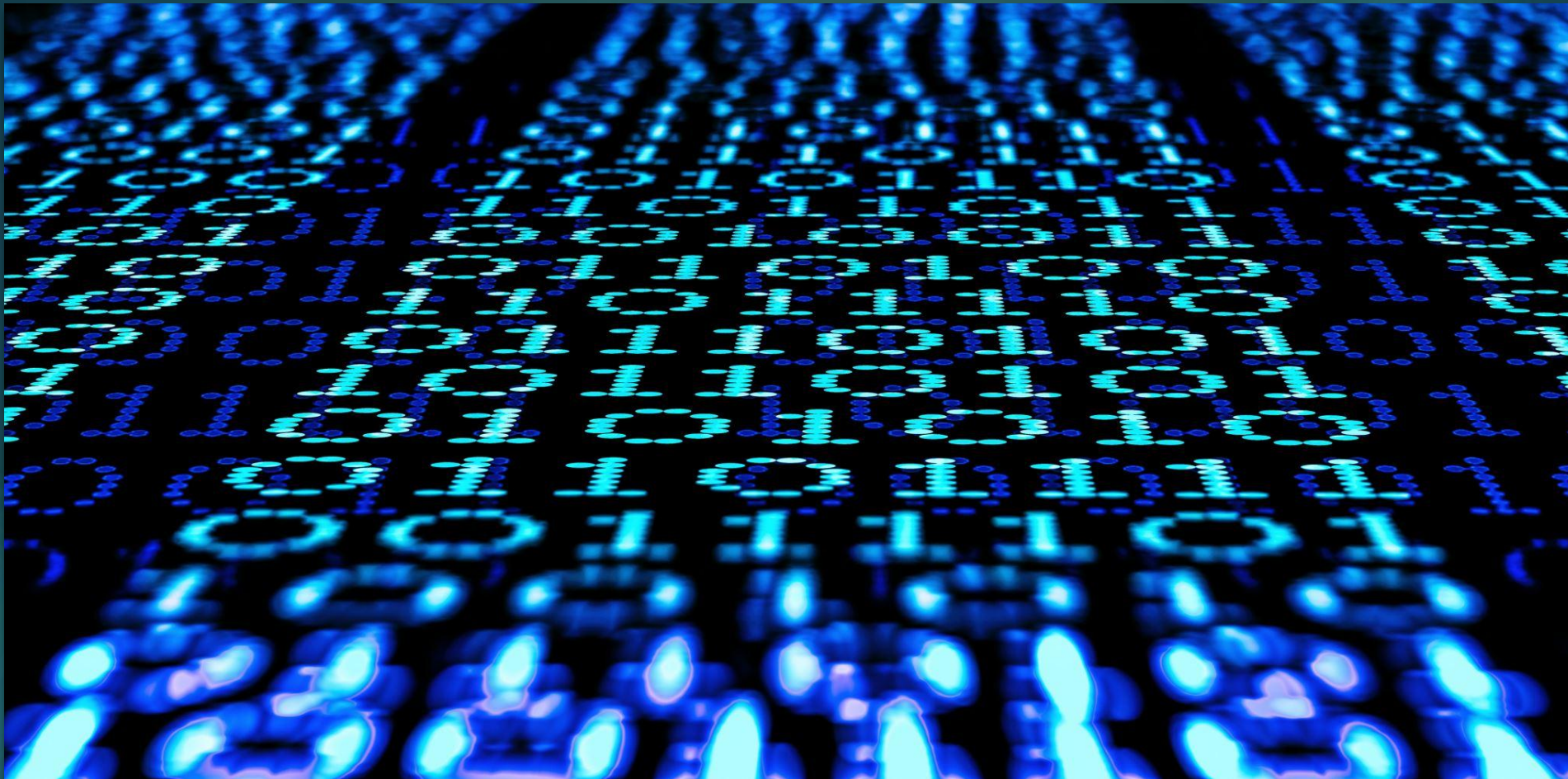




Fable Data Assignment (xx-06-2026)

“Classification of Merchant Descriptions using SQL-Server, Regex,
and Machine Learning”

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The “Why” and “How”

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Problem & Approach

Regex Baseline

Machine Learning Model

The “Other” Class

Recommendations & Future Enhancements

Q & A



Problem & Approach

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► The task:

- Classify merchant descriptions into b&m, evans, other

► Why it matters:

- ambiguity in merchant names, need for automation

► Workflow:

- SQL → Regex → ML → Clustering

Regex baseline: **Key points:**

SQL exploration revealed stable lexical patterns
Regex rules achieved **99.44%** accuracy

Strength: interpretable, fast

Limitation: brittle to unseen variations

REGEX BASELINE (THE “SURPRISINGLY STRONG” START)

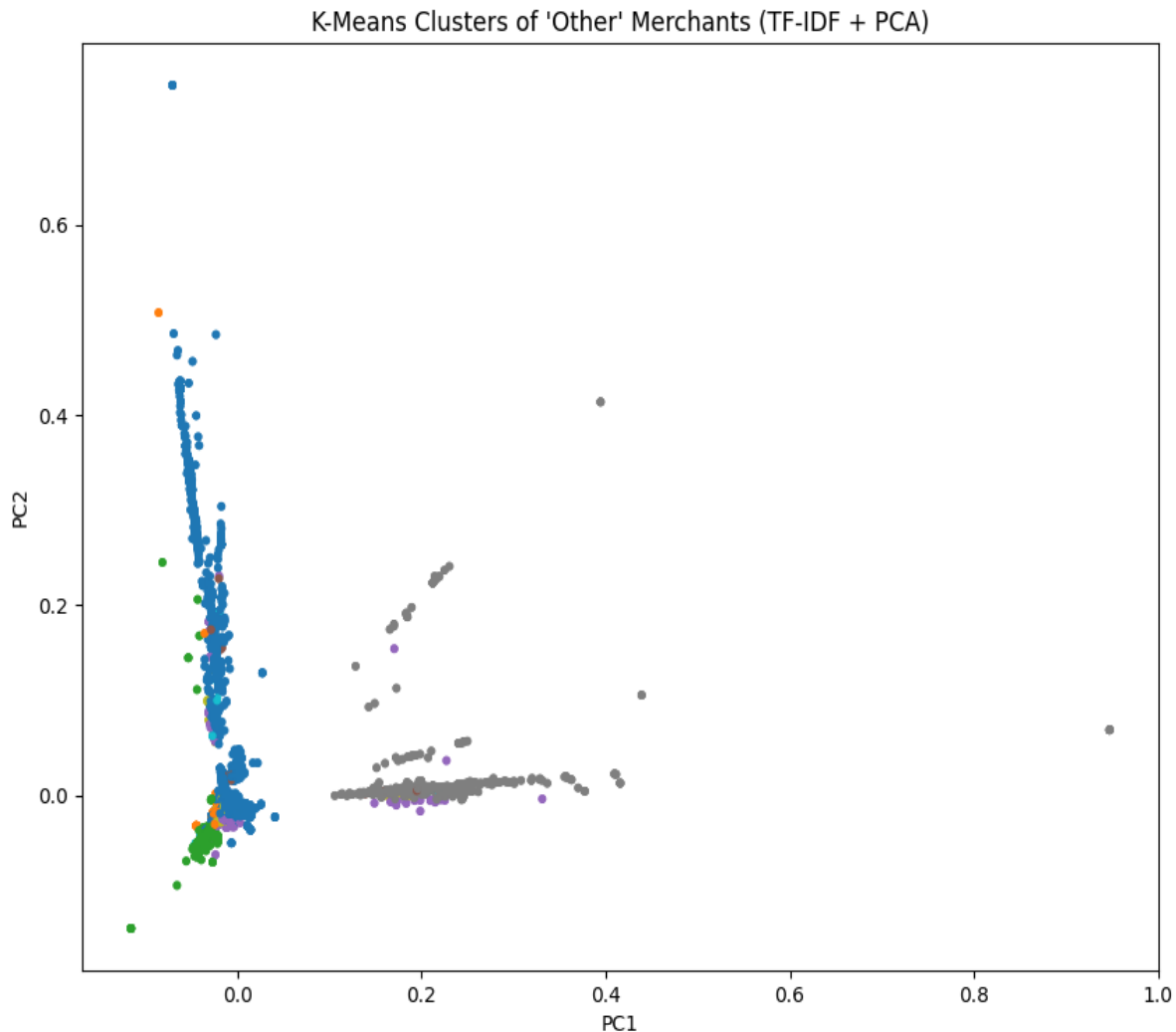
Machine Learning Model (The “Best Performer”)

- ▶ **TF-IDF/LR:** Test accuracy **99.56%**
- ▶ Captures subtle variations regex can't
- ▶ Interpretable coefficients
- ▶ Minimal overfitting

Model	Train Acc	Test Acc	Notes
Regex baseline	~99.0%	99.44%	Simple, interpretable, strong baseline
TF-IDF + Logistic Regression	99.97%	99.56%	Slightly better; more flexible and robust

Clustering the “Other” Class (The “Diagnostic Check”)

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- ▶ K-Means + PCA used to explore structure
- ▶ Clusters reflect naming conventions, not hidden merchant families
- ▶ Confirms “other” is heterogeneous, not a missing category

Model Comparison Summary

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Model	Purpose	Strengths	Weaknesses	Test Accuracy	Notes
Regex Baseline	Rule-based classifier derived from SQL pattern analysis	Extremely fast; interpretable; high precision for explicit name matches	Misses subtle or indirect patterns; brittle to spelling variation	99.44%	Strong baseline; demonstrates value of SQL exploration
TF-IDF + Logistic Regression	Main supervised model for classification	Best overall performance; robust; interpretable coefficients; handles variation well	Slightly more complex than regex; requires vectorisation	99.55%	Top performer ; perfect recall for both merchant classes
Multinomial Naïve Bayes	Simple generative baseline for comparison	Very fast; good with sparse text; easy to explain	Lower precision than LR; assumes feature independence	~97–99% (expected) 99.48% (actual)	Useful for contrasting discriminative models
Random Forest / Gradient Boosting	Tree-based comparison model	Captures non-linear patterns; good for tabular data	Expected to perform poorly on sparse TF-IDF; slower; higher memory use	~80–92% (expected) 98.70% (actual)	Exceeded expectations. Demonstrates linear models' ability to outperform trees for text processing.
K-Means + PCA (Clustering)	Explore structure of the “other” class	Reveals naming-pattern clusters; validates class heterogeneity	Not a classifier; no accuracy metric.	N/A	Confirms “other” not a hidden merchant category

Recommendations & Future Enhancements (“Forward Look”)

▶ **Short-term enhancements**

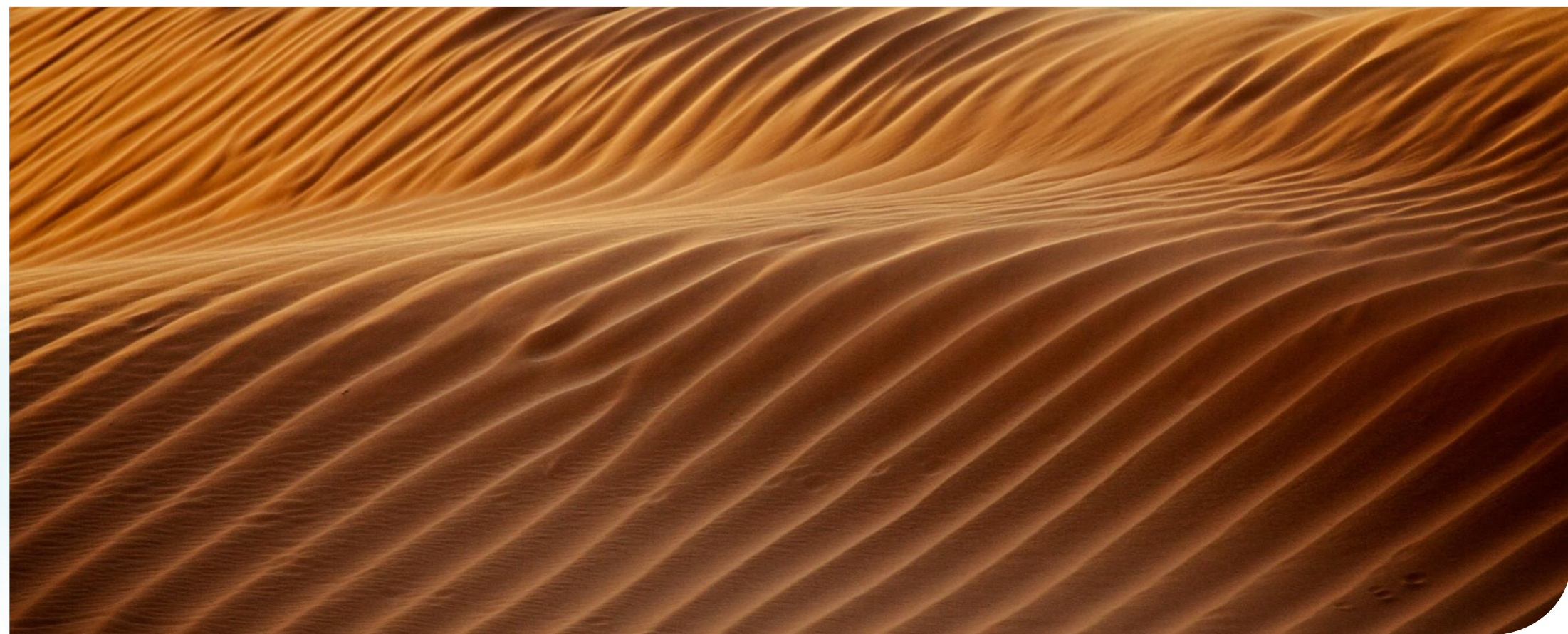
- ▶ Add unit tests for regex rules
- ▶ Deploy the LR model as a lightweight API
- ▶ Add monitoring for drift in merchant naming patterns

▶ **Medium-term enhancements**

- ▶ Add character-level models for extreme misspellings
- ▶ Explore semi-supervised learning if new merchant categories emerge
- ▶ Build a dashboard showing classification confidence and cluster drift

▶ **Long-term possibilities**

- ▶ Integrate with transaction metadata (amount, MCC codes, geography)
- ▶ Use embeddings only if text length increases or semantics matter



Q & A

Thank you for your attention.